BUILD

1.0.0 (Build 2)

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What is CTL?

CTL (Car Technical Log) is an aviation-style logbook application for car owners and fleet managers. Inspired by the rigorous maintenance records kept in aviation, CTL brings the same discipline to personal vehicle ownership — enabling users to track every maintenance event, parts replacement, fuel fill-up, insurance policy, and dealership visit for one or more vehicles.

The app is designed for drivers who want complete visibility into their vehicle's history, upcoming service schedule, and total cost of ownership. It supports multiple vehicles per account and generates comprehensive PDF-style reports per vehicle.

Core Value Proposition

- Complete vehicle history — every service, part, inspection, and repair in one place
- Upcoming event intelligence — never miss an oil change, tyre rotation, or MOT again
- Multi-vehicle support — manage your entire fleet from a single account
- Cost tracking — full breakdown of maintenance spend per vehicle
- Professional reports — shareable per-car logbook reports
- Offline-capable — local storage with cloud sync via secure API

Target Users

- Individual car owners who take maintenance seriously
- Enthusiasts tracking mods, servicing and inspection history
- Small fleet owners managing multiple company vehicles
- Used car buyers wanting to document a vehicle's full history

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High-Level Architecture

CTL follows a standard client-server architecture. The Expo mobile app communicates with a dedicated Express REST API over HTTPS. All persistent data lives in a PostgreSQL database managed via Drizzle ORM. Authentication uses JWT tokens stored securely on-device using expo-secure-store.

Architecture Diagram (Logical)



Technology Stack



Project Monorepo Structure



```
%    % % %  api-client-react/  Generated React Query hooks
```

```
% % %  pnpm-workspace.yaml
```

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Navigation Structure

CTL uses Expo Router's file-based routing with a tab bar at the bottom. The main entry points are the Login and Sign Up tabs. Once authenticated, users are navigated to the Home Dashboard which acts as the central hub for all features.

Tab Bar (Unauthenticated)



Main Navigation Stack (Post-Login)



Report

/report/[id]

Full logbook report per vehicle

[redacted]

Report List

/report/index

Choose vehicle for report

[redacted]

Log Event

/event/add

Quick-log any event type

Settings

/settings

Theme, notifications, account, sign out

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Feature Set Detail

Vehicle Management

- Add unlimited vehicles with: make, model, year, color, VIN, license plate, current mileage, photo
- Edit any vehicle details at any time
- Per-vehicle tabs: Maintenance · Parts · Insurance · Dealerships · Fuel
- Delete vehicle with all associated records

Maintenance Tracking

- Log service records: type, date, mileage at service, cost, shop name, notes
- Set next due date and/or next due mileage for scheduling future service
- Edit or delete existing maintenance records
- View upcoming events across entire fleet with urgency color-coding

Parts Tracking

- Record replaced parts: name, brand, category, part number, cost, supplier, warranty date
- Track warranty expiry and supplier contact

Insurance Management

- Log insurance policies: provider, policy number, type, premium, start/end date, coverage notes
- Track renewal dates alongside service schedule

Dealership Visits

- Record dealership/service centre visits: name, location, purpose, cost, contact, rating (1–5 stars)
- Track which shop handled which work

Fuel Log

- Log fill-ups: litres, cost per litre, total cost, mileage, fuel type, station name
- Track fuel economy trends over time

Malfunctions & Faults

- Record fault codes (OBD-II), warning lights, symptoms
- Log resolution status and repair cost
- Fleet-wide fault view and per-vehicle fault history

Inspections

- Schedule and record vehicle inspections (MOT, Roadworthy, etc.)
- Multiple inspection types supported
- Pass/fail outcome with notes

Reports

- Per-vehicle full logbook report with complete event timeline
- Cost breakdown by category (maintenance, parts, fuel, insurance, repairs)
- Upcoming service summary
- Sharable as PDF via native share sheet

Upcoming Events Dashboard

- Cross-fleet view of all events with upcoming due dates
- Urgency-coded by days remaining:

Ø=Ý4	Overdue	(past due date)
Ø=ßà	Urgent	("d 7 days)
Ø=ßá	Approaching	("d 30 days)
Ø=ßâ	Scheduled	(> 30 days)

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Authentication Flow

CTL uses a standard JWT-based authentication system. Credentials never leave the device in plain text — passwords are hashed with bcryptjs (salt rounds: 12) server-side before storage. Tokens are stored in expo-secure-store which uses the iOS Keychain and Android Keystore.

Sign Up

- User submits name, email, and password
- Server hashes password with bcryptjs and stores user record
- Returns signed JWT (expires: 7 days)
- App stores token in secure storage and navigates to Home

Sign In

- User submits email + password
- Server compares with bcrypt hash
- On success: returns JWT; on failure: 401 Unauthorized
- After first successful login, user is prompted to enable biometric sign-in

Biometric Sign-In (Face ID / Touch ID / Fingerprint)

- Available on devices with enrolled biometrics
- Stored JWT is retrieved from secure storage and validated via biometric prompt
- No credentials are transmitted during biometric flow — only the stored token
- User preference persisted under key: ctl_biometric_enabled

Token Management



Security Considerations

- All API requests require a valid Bearer token in the Authorization header
- Server validates token signature and expiry on every protected endpoint
- Tokens expire after 7 days; users must re-authenticate after expiry
- Password reset uses time-limited single-use tokens (future: email delivery)
- Account deletion permanently removes all user data and associated vehicle records
- HTTPS enforced in production; all communication is TLS-encrypted

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Database Schema Overview

CTL uses a PostgreSQL relational database managed through Drizzle ORM. All tables use integer serial primary keys with cascading deletes so that removing a user removes all their vehicles, and removing a vehicle removes all associated records.

users



cars



maintenance_records



date

date

Date of service

[redacted]

mileage

integer

Odometer at service

cost

numeric(10,2)

Service cost

shop

varchar(200)

Service centre name

next_due_date

date

When next service is due

next_due_mileage

integer

Mileage trigger for next service

created_at

timestamp

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[redacted]

parts_records

[redacted]

insurance_records

[redacted]

dealership_records

[redacted]

date

date

Visit date

[redacted]

cost

numeric(10,2)

Total cost

contact

varchar(200)

Contact phone/email

rating

integer

1–5 star rating

notes

text

Optional notes

fuel_records



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[redacted]

Base URL & Authentication

Base URL (dev):	http://localhost:8080
Base URL (prod):	https://[deployed-domain].replit.app
Auth header:	Authorization: Bearer <jwt_token>
Content-Type:	application/json

[redacted]

Authentication Endpoints

[redacted]

Vehicle (Cars) Endpoints

[redacted]

Maintenance Endpoints

[redacted]

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[redacted]

Parts Endpoints

[redacted]

Insurance Endpoints

[redacted]

Dealership Endpoints

[redacted]

Fuel Endpoints

[redacted]

Malfunctions & Inspections

[redacted]

GET

/malfunctions

Yes

All malfunctions (fleet-wide)

[redacted]

GET

/cars/:carId/malfunctions

Yes

Malfunctions for one vehicle

POST

/cars/:carId/malfunctions

Yes

Log new malfunction

DELETE

/cars/:carId/malfunctions/:recordId

Yes

Delete malfunction

GET

/inspections

Yes

All inspections (fleet-wide)

GET

/cars/:carId/inspections

Yes

Inspections for one vehicle

POST

/cars/:carId/inspections

Yes

Log new inspection

DELETE

/cars/:carId/inspections/:recordId

Yes

Delete inspection

Events & Reports



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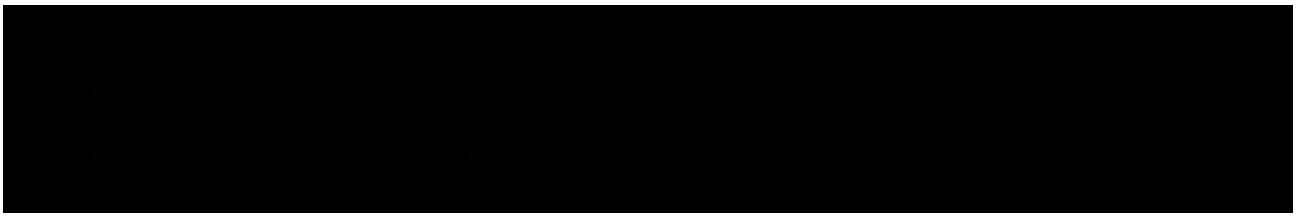
Overview

CTL uses expo-notifications (v0.32.x) to deliver local push notifications — no third-party push notification server is required. Notifications are scheduled on-device each time the app comes to the foreground, based on the current state of upcoming events fetched from the API.

Notification Types

Weekly Digest (Every Sunday at 9:00 AM)

A summary notification is scheduled for the next Sunday at 9 AM whenever the app is opened. It lists up to 5 upcoming events sorted by urgency (most critical first).



Individual Event Reminders

Separate notifications are scheduled at three thresholds before each event's due date:



Implementation Details

- Permission requested on first app open after login (iOS and Android 13+)
- All existing scheduled notifications are cancelled and rescheduled on every app foreground
- Past thresholds (already overdue or past the reminder date) are silently skipped
- Notifications fire at 9:00 AM local device time
- Web platform is excluded (notifications only on native iOS and Android)

Key Files



Required Permissions



iOS

NSUserNotificationUsageDescription

System permission dialog

Android

POST_NOTIFICATIONS

Android 13+ required

Android

SCHEDULE_EXACT_ALARM

Precise alarm scheduling

Android

RECEIVE_BOOT_COMPLETED

Persist notifications after reboot

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Overview

CTL ships with a full Light / Dark / System theme system. The user's preference is persisted on-device via expo-secure-store and applied instantly without an app restart. All 42 screens and UI components react to theme changes in real time.

Theme Options



Architecture

- ThemeContext.tsx — React context providing useColors() hook and useTheme() hook
- colors.ts — Defines the full light and dark ColorScheme with named semantic tokens
- useColors() — Returns the active ColorScheme based on current preference + OS setting
- useTheme() — Exposes { preference, setPreference, resolvedTheme }
- Preference stored under SecureStore key: ctl_theme_preference

Color Tokens



Settings UI

The Settings screen exposes a 3-option segmented control with sun/moon/smartphone icons for Light, Dark,

and System respectively. Tapping a segment triggers haptic feedback and immediately applies the new theme across the entire app.

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Build System

CTL uses EAS (Expo Application Services) for native builds. Builds are triggered via the EAS CLI and can be submitted directly to TestFlight (iOS) or the Google Play Console (Android).

Build Configuration

EAS Account:	mkashour
iOS Bundle ID:	com.ctl.carlogbook
Android Package:	com.ctl.carlogbook
iOS Build Number:	2 (current TestFlight)
App Version:	1.0.0
Build Profile:	preview

Build Commands

iOS build

Android build

iOS submit

Distribution Status

API Server Deployment

The Express API server is deployed on Replit. For production, it should be promoted to a dedicated always-on deployment to avoid cold start latency. The DATABASE_URL environment variable points to the provisioned PostgreSQL instance.

- Current: Replit development server (may sleep on inactivity)
- Recommended: Promote to Replit Deployment for 24/7 uptime
- Database: Replit-managed PostgreSQL (persistent across restarts)

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[redacted]

Current Limitations



Development Roadmap

Immediate (Next Build)

- Password reset via email (SendGrid or Resend integration)
- Deploy API server to always-on production environment
- App Store screenshots for iOS submission

Short Term

- Onboarding flow for first-time users
- Edit Profile screen (name, email, password change)
- Mileage unit toggle (km !” miles) with full conversion throughout app
- In-app feedback form

Medium Term

- Push notifications via Expo Push Notification Service (cloud-triggered reminders)
- Calendar integration — sync upcoming events to device calendar
- Fuel economy charts and trend analysis
- Full-text search across all records
- Cloud photo storage for vehicle and receipt photos

Long Term / CTL Fleet

- Multi-user / team accounts for fleet managers
- Advanced analytics dashboard — cost per km, lifecycle reporting
- Workshop / mechanic portal
- OBD-II Bluetooth scanner integration for automatic fault code logging
- Insurance renewal reminders with provider comparison